

AIAA 4432: Statistical Machine Learning

Yao SHU

AI Thrust, HKUST (GZ)

2026.09.03 · Fall 2026/27

Outline

1. Why a Theory Course, Now
2. Risk, Overfitting, and What a Training Score Can Say
3. Finite-Class Generalization

Part I · Foundations · Week 1

Section 1: Why a Theory Course, Now

Course logistics, learning outcomes, assessment — and what experiments alone cannot tell you

Course Section and Contacts

Course

AIAA 4432· Statistical Machine Learning
3 credits · Fall 2026/27
English

Section

Thursday 12:00–14:50 · E1147

Prerequisites

DSAA 2011 and UFUG 2104

Instructor

Dr. Yao
yaoshu@hkust-gz.edu.cn
<https://yao.notion.site>

Teaching assistants

QIU Junbin
jqiu236@connect.hkust-gz.edu.cn
LIN Hongbin
hlin199@connect.hkust-gz.edu.cn

Office hour

Thursday 15:30–17:30

Course Learning Outcomes

1 Explain and derive
statistical principles behind representative methods.

2 Implement and evaluate
methods with a valid evaluation protocol.

3 Select and critique
assumptions, data regimes, uncertainty, and failures.

4 Communicate
claims, evidence, limitations, and responsible use.

Assessment Structure: Criterion-Referenced

Component	Weight	Timing	What it measures / CILOs
Learning Checks	10%	every class; best 10 of 12	recall + confidence / 1–3
Quizzes (In-class)	25%	W2, 4, 6, 8, 10, 12; best 5 of 6	short reasoning + diagnosis / 1, 3
Individual Project	25%	one package, W9–12	analysis + code + video / 2–4
Final Examination	40%	end of term	concepts · derivation · counterexample · interpretation / 1, 3, 4
Total	100%		

Project late rule: days 1–3, –30%; days 4–5, –60%; days 6–7, no credit; after day 7, not accepted.

GenAI: disclose its role and verify claims, proofs, code, citations, data, and results. No GenAI in the Final.

Read This Before You Decide to Stay

This course is hard, in a specific way.

- You will rebuild proofs from named steps, not merely recognize them.
- The Final (40 %) includes unseen methods. Credit follows valid reasoning under stated assumptions, not a memorized formula used in the wrong place.
- Proofs require steady weekly work; they do not compress into the night before a quiz.

This is the first offering of this design.

- Materials may be revised during the term; Canvas is the definitive source.
- Corrections and suggestions are welcome and never affect marks.

If you need a predictable workload and a safe GPA, this is probably not the course.
Better to know before the add/drop deadline.

The Empirical Case Is Already Won

Modern ML works.

Large models trained by gradient descent on enormous data beat almost every hand-designed method.

The Empirical Case Is Already Won

Modern ML works.

Large models trained by gradient descent on enormous data beat almost every hand-designed method.

So why spend a term on theorems?

The Empirical Case Is Already Won

Modern ML works.

Large models trained by gradient descent on enormous data beat almost every hand-designed method.

So why spend a term on theorems?

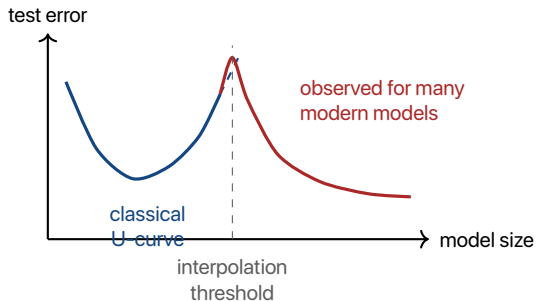
Because you already follow rules nobody derived for you; you already read numbers nobody qualified for you; and at least one of those rules is not even true.

Rules You Already Follow — Without Being Told Why

Rule from your intro ML course	Question it leaves open	Week
Never score on the training set.	Why is that score optimistic? By how much?	W1
Hold out a test set; touch it once.	What does each extra look cost?	W1, W8
More data helps.	How much? Does the model class matter?	W1, W3, W4
Regularize, early-stop, or shrink the model.	What is the common mechanism? When does "smaller" hurt?	W5–W7
Test on the training distribution.	What if deployment differs? Can we correct the score?	W8

Each rule has a reason and a boundary. This course states both.

Where the Rule You Were Taught Stops Being True



Old rule: past some size, a bigger model overfits.

Observation (Belkin et al. 2019; Nakkiran et al. 2020): after interpolation, test error can **fall again**.

Preview (Weeks 6–7): adding features can make a lower-norm interpolant available. In kernel/NTK-like regimes, implicit bias may select it: parameter count rises while effective complexity falls.

Missing term: class size ignores which solution the algorithm selects. More parameters can enable a lower-norm interpolant; Weeks 6–7 state the scope.

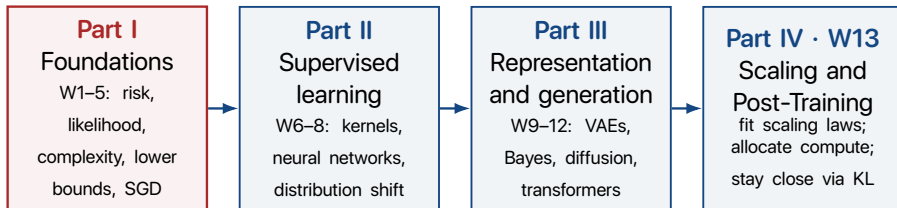
From a Plot to a Result

The double-descent plot is an observation, not yet an explanation. To explain why the curve turns down, a theoretical result must state five things:

- 1 **Object** — which quantity is being bounded: a risk, a divergence, a sample size.
- 2 **Assumptions** — i.i.d. data? realizable? bounded loss? which model class?
- 3 **Statement** — an inequality with explicit constants or rates.
- 4 **Probability** — over what randomness, with what confidence.
- 5 **Scope** — where it applies and where it is known to fail.

One Question, Thirteen Weeks

The course question. When a model is chosen using data, why should it work on data it has not seen — and what does that choice cost in samples and computation?



Names ahead: generalization and sample complexity; approximation and estimation error; stability; minimax lower bounds.

Course shorthand: price means a bound induced by a restriction; **A / E / O** means approximation / estimation / optimization. Standard terms remain primary.

What You Will Be Able to Do by Week 13

Given a method you have **never seen**:

- write down what it estimates, and under which loss;
- name the assumption that lets a data-chosen output generalize;
- state the resulting bound, its rate, and what it depends on;
- design an experiment that could show the bound is loose or the assumption false;
- say clearly what the analysis does **not** cover.

The Final's four categories — concepts · derivation · counterexample · interpretation — are these five bullets.

Part I · Foundations · Week 1

Section 2: Risk and Overfitting

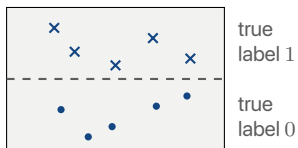
From a familiar failure to its exact cause — then names for everything involved

A Classifier You Have Already Met

Data: unit-square points; label 1 above the dashed line, 0 below. Train on 200 points.

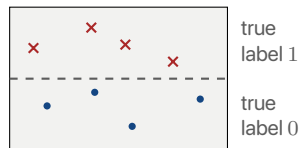
Memorizer: return a stored point's label; **return 0 everywhere else.**

Scored on the training set



stored points return their labels
200/200 correct

Scored on 1,000 fresh points



every fresh point gets prediction 0
below: **correct**; above: **wrong**

● label 0 × label 1 blue = correct red = wrong

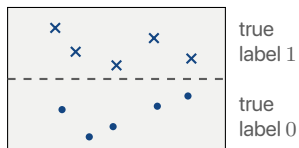
training accuracy **100 %**

A Classifier You Have Already Met

Data: unit-square points; label 1 above the dashed line, 0 below. Train on 200 points.

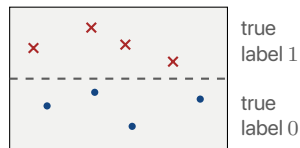
Memorizer: return a stored point's label; **return 0 everywhere else.**

Scored on the training set



stored points return their labels
200/200 correct

Scored on 1,000 fresh points



every fresh point gets prediction 0
below: **correct**; above: **wrong**

● label 0 × label 1 blue = correct red = wrong

training accuracy **100 %**

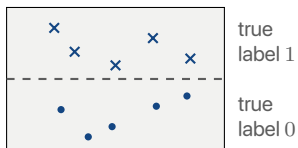
fresh-data accuracy $\approx$ **50 %**

A Classifier You Have Already Met

Data: unit-square points; label 1 above the dashed line, 0 below. Train on 200 points.

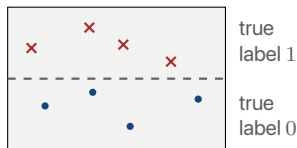
Memorizer: return a stored point's label; **return 0 everywhere else.**

Scored on the training set



stored points return their labels
200/200 correct

Scored on 1,000 fresh points



every fresh point gets prediction 0
below: **correct**; above: **wrong**

● label 0 × label 1 blue = correct red = wrong

training accuracy **100 %**

fresh-data accuracy $\approx$ **50 %**

Why?

Name It Precisely

The everyday name for what just happened is **overfitting**:

- ① On a training point, the memorizer copies that point's label; its 100 % score was settled before we looked.
- ② On a fresh point it has nothing to copy, so it predicts 0; fresh accuracy is $\frac{1}{2}$.
- ③ Thus the training score never estimated the fresh score. **The same data built and scored the rule.**

Precise statement. A score on the sample used to build or select a rule need not estimate its fresh-data accuracy.

Why "select". Among 2^{200} labelings, the sample selects one matching all 200 points. Best-fit selection creates optimism; more candidates allow more. Section 3 quantifies it.

Overfitting is not a property of a model. It is what happens when one sample does two jobs: choosing the rule and scoring it.

Training a Model Is Choosing from a Class

Empirical risk minimization (ERM): from a class $\mathcal{H}$ of candidate rules, keep the one with the **lowest training error**. Most methods you know — logistic regression, trees, neural networks — are versions of this.

The memorizer		Supervised learning in general	
all 2^{200} labelings of the training points	$\longleftrightarrow$	the hypothesis class $\mathcal{H}$	
the 200 stored points	$\longleftrightarrow$	the training set S	
pick the labeling that matches the sample	$\longleftrightarrow$	ERM: minimize training error	
the fresh points	$\longleftrightarrow$	test data / the population	
100 % on the sample vs. $\frac{1}{2}$ on fresh points	$\longleftrightarrow$	the generalization gap of the learned model	

Learning selects from a class using S . The generalization gap depends on how much choice the class permits.

Objects in a Learning Problem



Population law

$(X, Y) \sim \mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$; $\mathcal{D}$ is unknown.

Sample

$S = ((x_1, y_1), \dots, (x_n, y_n)) \sim \mathcal{D}^n$.

Hypothesis class

$\mathcal{H}$ is the set of predictors we allow — the k candidates.

Algorithm

A uses S to select $h_S = A(S) \in \mathcal{H}$.

i.i.d. is an assumption about data collection—not a fact created by shuffling a table.

Population and Empirical Risk

True risk (generalization error)

$$L_{\mathcal{D}}(h) = \mathbb{P}_{(x,y) \sim \mathcal{D}}[h(x) \neq y].$$

fixed but unavailable

Empirical risk (training error)

$$L_S(h) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[h(x_i) \neq y_i].$$

observable but random

A score of 100% may describe the observed items exactly and still fail to predict fresh items.

What Is Random in a Probability Statement?

Expression	What is random	Question answered
$\mathbb{P}_{(X,Y) \sim \mathcal{D}}$	a fresh test point	How often does a fixed rule fail?
$\mathbb{P}_{S \sim \mathcal{D}^n}$	a fresh training set	How often is the learned rule bad?
$\mathbb{P}(Y = 1 \mid X = x)$	the label at fixed x	How uncertain is the label here?

For learned h_S , take the test draw independently of S ; the two layers coexist:

$$\mathbb{P}_{S \sim \mathcal{D}^n} \left(\underbrace{\mathbb{P}_{(X,Y) \sim \mathcal{D}}[h_S(X) \neq Y]}_{L_{\mathcal{D}}(h_S)} > \varepsilon \right).$$

Read the subscript of $\mathbb{P}$ first: it tells you which randomness is being considered.

Indicator Expectations and Conditioning

For any event E ,

$$\mathbb{E}[\mathbf{1}[E]] = \mathbb{P}(E).$$

For any integrable random variable U , the tower property says

$$\mathbb{E}[U] = \mathbb{E}_X[\mathbb{E}[U \mid X]].$$

With $U = \mathbf{1}[h(X) \neq Y]$,

$$L_{\mathcal{D}}(h) = \mathbb{E}_X[\mathbb{P}(Y \neq h(X) \mid X)].$$

Indicator $\rightarrow$ probability; conditioning $\rightarrow$ pointwise problem; outer expectation $\rightarrow$ population risk.

The Best Possible Rule

Before analysing learned h_S , ask: what is the **smallest** risk any rule can attain?

Assumptions.

- Binary $Y \in \{0, 1\}$, 0–1 loss, and joint law $(X, Y) \sim \mathcal{D}$.
- $\eta(x) = \mathbb{P}(Y = 1 \mid X = x)$. Values 0 or 1 mean a determined label; $\frac{1}{2}$ means maximal uncertainty.
- No sample S : every statement concerns $\mathcal{D}$ alone.

Starting point (tower property):

$$L_{\mathcal{D}}(h) = \mathbb{E}_X \left[\underbrace{\mathbb{P}(Y \neq h(X) \mid X)}_{\text{conditional error at } X} \right].$$

Plan: minimize conditional error at each x , then average. A function problem becomes two choices per x .

The Conditional Error Is a Two-Way Choice

Fix x and abbreviate $\eta = \eta(x)$. A rule can only do one of two things at x :

$$\mathbb{P}(Y \neq 0 \mid X = x) = \mathbb{P}(Y = 1 \mid X = x) = \eta,$$

$$\mathbb{P}(Y \neq 1 \mid X = x) = \mathbb{P}(Y = 0 \mid X = x) = 1 - \eta.$$

decision at x	conditional error	it is the smaller one when
$h(x) = 1$	$1 - \eta$	$\eta \geq \frac{1}{2}$
$h(x) = 0$	η	$\eta \leq \frac{1}{2}$

Hence, for every rule h and every x ,

$$\mathbb{P}(Y \neq h(x) \mid X = x) \geq \min\{\eta(x), 1 - \eta(x)\},$$

with equality when $h(x) = \mathbf{1}[\eta(x) \geq \frac{1}{2}]$ (either choice ties at $\eta = \frac{1}{2}$).

This inequality holds pointwise, before any expectation is taken. That is what makes the next step a one-liner.

Theorem: The Bayes Classifier (0–1 Loss)

Theorem (Bayes optimal predictor)

Let $Y \in \{0, 1\}$, $(X, Y) \sim \mathcal{D}$, and measurable $\eta(x) = \mathbb{P}(Y = 1 \mid X = x)$. Define $f^*(x) = \mathbf{1}[\eta(x) \geq \frac{1}{2}]$. For every measurable $h : \mathcal{X} \rightarrow \{0, 1\}$,

$$L_{\mathcal{D}}(f^*) \leq L_{\mathcal{D}}(h), \quad \text{and} \quad L_{\mathcal{D}}(f^*) = \mathbb{E}_X[\min\{\eta(X), 1 - \eta(X)\}].$$

These are population quantities: no sample or confidence level.

Proof. Pointwise, $\mathbb{P}(Y \neq h(x) \mid X = x) \geq \min\{\eta(x), 1 - \eta(x)\} = \mathbb{P}(Y \neq f^*(x) \mid X = x)$. Monotonicity and the tower property give

$$L_{\mathcal{D}}(h) = \mathbb{E}_X[\mathbb{P}(Y \neq h(X) \mid X)] \geq \mathbb{E}_X[\min\{\eta, 1 - \eta\}] = L_{\mathcal{D}}(f^*). \quad \square$$

Scope. f^* depends on unknown $\mathcal{D}$: it is a benchmark, not a data-built rule. Binary labels and 0–1 loss only.

In-Class Exercise: 90 seconds

At some x , $\eta(x) = 0.72$. What does f^* predict, and what is the conditional error there? What if $\eta(x) = 0.5$?

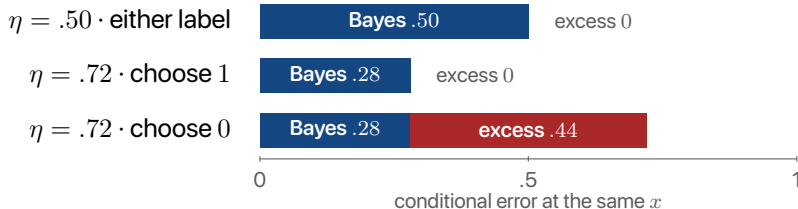
In-Class Exercise: 90 seconds

At some x , $\eta(x) = 0.72$. What does f^* predict, and what is the conditional error there? What if $\eta(x) = 0.5$?

$\eta = 0.72$: predict 1; the two candidate errors are 0.72 and 0.28; the smaller, 0.28, remains and no rule can do better at that x . $\eta = 0.5$: either prediction; error 0.5 either way.

Bayes Risk Is the Floor; Excess Risk Is the Avoidable Gap

$$\underbrace{L_{\mathcal{D}}(h)}_{\text{total population risk}} = \underbrace{L_{\mathcal{D}}(f^*)}_{\text{Bayes risk irreducible floor}} + \underbrace{\mathcal{E}(h)}_{\text{excess risk avoidable gap}}, \quad \mathcal{E}(h) := L_{\mathcal{D}}(h) - L_{\mathcal{D}}(f^*) \geq 0.$$



Target: excess over the Bayes benchmark.

Next: $\mathcal{D}$ is unknown — when does the computable $L_S(h)$ estimate $L_{\mathcal{D}}(h)$?

What We Can Compute, and What We Want

For a rule h , the target is $L_{\mathcal{D}}(h)$, or its excess over Bayes risk.

Cannot be computed:

$$L_{\mathcal{D}}(h) = \mathbb{P}_{(X,Y) \sim \mathcal{D}}[h(X) \neq Y].$$

A fixed number, but $\mathcal{D}$ is unknown.

Can be computed:

$$L_S(h) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[h(x_i) \neq y_i].$$

Random because it depends on S .

Question: when does computable $L_S(h)$ estimate $L_{\mathcal{D}}(h)$?

Two cases behave differently:

Case 1 h was fixed before drawing S .

Case 2 h was chosen using S .

Case 1: a variance calculation works. Case 2: the next slides identify exactly where it breaks.

Case 1: A Hypothesis Fixed Before the Data

Let h be fixed, not depending on S . Put $Z_i = \mathbf{1}[h(X_i) \neq Y_i]$, so $L_S(h) = \frac{1}{n} \sum_i Z_i$.

- 1 **Unbiased.** Since $(X_i, Y_i) \sim \mathcal{D}$ and h is fixed, $\mathbb{E}[Z_i] = L_{\mathcal{D}}(h)$ and $\mathbb{E}[L_S(h)] = L_{\mathcal{D}}(h)$.
- 2 **Variance.** With $p = L_{\mathcal{D}}(h)$, each Bernoulli Z_i has variance $p(1-p)$. The Z_i are independent because they use independent pairs through the same fixed h :

$$\text{Var}(L_S(h)) = \frac{1}{n^2} \sum_i \text{Var}(Z_i) = \frac{p(1-p)}{n} \leq \frac{1}{4n}.$$

- 3 **Concentration.** Chebyshev gives

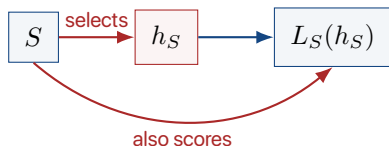
$$\mathbb{P}_{S \sim \mathcal{D}^n}(|L_S(h) - L_{\mathcal{D}}(h)| \geq \varepsilon) \leq \frac{1}{4n\varepsilon^2}.$$

Week 3: Hoeffding sharpens $1/(4n\varepsilon^2)$ to $2e^{-2n\varepsilon^2}$.

" h is fixed" licenses both $\mathbb{E}[Z_i] = L_{\mathcal{D}}(h)$ and independence of the Z_i . The next slide breaks both.

Case 2: A Hypothesis Chosen With the Data

Empirical risk minimization picks $h_S \in \arg \min_{h \in \mathcal{H}} L_S(h)$: the best candidate in $\mathcal{H}$, judged on S .



Replace h by h_S : $Z_i = \mathbf{1}[h_S(X_i) \neq Y_i]$.

- 1' $\mathbb{E}[Z_i] = L_{\mathcal{D}}(h_S)$? **Not licensed**: h_S uses all of S , including (X_i, Y_i) ; $L_{\mathcal{D}}(h_S)$ is also random.
- 2' Independence of the Z_i ? **Lost**. All Z_i share the same h_S .
- 3' With 1' and 2' gone, Chebyshev has nothing to act on.

The observations remain i.i.d. What fails is **independence between the selected rule and the sample scoring it**: one sample, two jobs.

$\mathbb{E}_S[L_S(h_S)] = \mathbb{E}_S[L_{\mathcal{D}}(h_S)]$ is not guaranteed. For the memorizer, the next slide shows a gap of $\frac{1}{2}$ for every n .

Case 2, Worst Case: The Memorizer, Formally

Setting. X is uniform on $[0, 1]^2$; $Y = \mathbf{1}[X \in U]$ for the upper half U , so Bayes risk is 0. Let $\mathcal{H}$ contain all functions $\mathcal{X} \rightarrow \{0, 1\}$ and

$$h_S(x) = \begin{cases} y_i, & x = x_i \text{ for some } i \leq n, \\ 0, & \text{otherwise.} \end{cases}$$

Fact. Under a density, every finite set has probability 0 (Notes, Lemma C.2).

- ❶ $L_S(h_S) = 0$ by construction, so h_S is an ERM.
- ❷ On fresh x , it errs exactly when $x \in U \setminus \{x_1, \dots, x_n\}$. Hence

$$L_{\mathcal{D}}(h_S) = \mathbb{P}(X \in U, X \notin \{x_1, \dots, x_n\}) = \mathbb{P}(X \in U) - \underbrace{\mathbb{P}(X \in \{x_1, \dots, x_n\} \cap U)}_{=0} = \frac{1}{2}.$$

- ❸ Thus $\mathbb{E}_S[L_S(h_S)] = 0$ but $\mathbb{E}_S[L_{\mathcal{D}}(h_S)] = \frac{1}{2}$: the gap is $\frac{1}{2}$ **for every** n .

More data alone does not close the gap. We must restrict how much the sample may choose.

The Central Problem, in Words

Fixed h : $L_S(h)$ estimates $L_{\mathcal{D}}(h)$, with variance $\leq 1/(4n)$.

Data-chosen h_S : $L_S(h_S)$ may miss $L_{\mathcal{D}}(h_S)$ by $\frac{1}{2}$ for every n .

Learning theory restricts the choice and bounds the resulting gap.

Why can restriction help? The memorizer offers all 2^n sample labelings, so one always fits. For each fixed bad candidate, a perfect sample fit is luck; with few candidates, such luck is rare.

Section 3 proves: for finite $\mathcal{H}$, the optimism is controlled by $|\mathcal{H}|$ and n .

Excess-Risk Decomposition: Connecting the Target to ERM

Comparison rules. f^* is Bayes optimal; $h_{\mathcal{H}}^* \in \arg \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h)$ is best in $\mathcal{H}$; $\hat{h}_S$ is the ERM.

Add and subtract $L_{\mathcal{D}}(h_{\mathcal{H}}^*)$:

$$\underbrace{L_{\mathcal{D}}(\hat{h}_S) - L_{\mathcal{D}}(f^*)}_{\text{excess risk of ERM}} = \underbrace{L_{\mathcal{D}}(h_{\mathcal{H}}^*) - L_{\mathcal{D}}(f^*)}_{\mathbf{A}: \text{approximation error}} + \underbrace{L_{\mathcal{D}}(\hat{h}_S) - L_{\mathcal{D}}(h_{\mathcal{H}}^*)}_{\mathbf{E}: \text{estimation error}}.$$

- $\mathbf{A} \geq 0$: restriction to $\mathcal{H}$. It depends on $\mathcal{H}, \mathcal{D}$, not on the sample.
- $\mathbf{E} \geq 0$: sample-based choice within $\mathcal{H}$. Section 3 bounds it.
- $\mathbf{O} = L_{\mathcal{D}}(\tilde{h}_S) - L_{\mathcal{D}}(\hat{h}_S)$ measures failure to reach ERM. Today $\tilde{h}_S = \hat{h}_S$, so $\mathbf{O} = 0$.

Larger $\mathcal{H}$ can decrease $\mathbf{A}$ but increase the bound on $\mathbf{E}$. The trade-off is between $\mathbf{A}$ and a bound on $\mathbf{E}$.

The Decomposition on the Opening Example

Same $\mathcal{D}$ as before: $Y = \mathbf{1}[X \in U]$, upper half, Bayes risk 0, $f^* = \mathbf{1}[x_2 \geq \frac{1}{2}]$. Three choices of class:

Class $\mathcal{H}$	$h_{\mathcal{H}}^*$	A	E	why
{constant 0}	constant 0	$\frac{1}{2}$	0	one candidate: all error is approximation
all functions $\mathcal{X} \rightarrow \{0, 1\}$	f^*	0	$\frac{1}{2}$	memorizer is an ERM: all error is estimation
$\{\mathbf{1}[x_2 \geq t] : t \in [0, 1]\}$	$f^* (t = \frac{1}{2})$	0	small	$f^* \in \mathcal{H}$; few failure modes

A approximation

Can $\mathcal{H}$ express the target?

E estimation

How optimistic is $\hat{h}_S$?

O optimization

Did computation reach ERM?

Three Questions, Answered for Today

Every analysis asks three questions. Today's answers:

Q1 What restricts the choice?

Case 1: one fixed rule. All functions: no restriction. Section 3: finite $\mathcal{H}$.

Q2 Assumptions and bound?

Case 1: i.i.d.; gap $\leq \varepsilon$ w.p. $\geq 1 - 1/(4n\varepsilon^2)$.
Section 3: i.i.d. + realizable; gap
 $\leq \log(|\mathcal{H}|/\delta)/n$.

Q3 Which term?

Estimation error $\mathbf{E}$; restricting $\mathcal{H}$ may create approximation error $\mathbf{A}$.

Tools ahead: counting / concentration (today, W3) · stability (W5) · sample splitting (W5, W8) · lower bounds (W4).

Part I · Foundations · Week 1

Section 3: Finite-Class Bounds

A complete PAC proof under realizability, followed by its limitations

Realizability and the PAC Objective

Definition (Realizability)

There exists $h^* \in \mathcal{H}$ with

$$L_{\mathcal{D}}(h^*) = 0.$$

This implies Bayes risk zero:

$\eta(x) \in \{0, 1\}$ almost everywhere.

Definition (PAC learnability)

For every $\varepsilon, \delta \in (0, 1)$ and every realizable $\mathcal{D}$, there is $n_{\mathcal{H}}(\varepsilon, \delta)$ such that

$$\mathbb{P}_{S \sim \mathcal{D}^n}(L_{\mathcal{D}}(A(S)) \leq \varepsilon) \geq 1 - \delta.$$

whenever $n \geq n_{\mathcal{H}}(\varepsilon, \delta)$.

ε : accuracy δ : confidence $n_{\mathcal{H}}$:
sample complexity

The theorem below treats a finite realizable class. In general, Bayes risk need not be zero.

Three Probability Rules the Proof Will Use

- 1 **Complement:** $\mathbb{P}(E^c) = 1 - \mathbb{P}(E)$.
- 2 **Independent repetition:** $\mathbb{P}(\text{miss } E \text{ in all } n \text{ draws}) = (1 - \mathbb{P}(E))^n$.
- 3 **Union bound:** $\mathbb{P}\left(\bigcup_{j=1}^m E_j\right) \leq \sum_{j=1}^m \mathbb{P}(E_j)$.

Example (the multiple-testing row of Section 2): if each of 20 independent tests reports $p < .05$ with probability .05, then

$$\mathbb{P}(\text{at least one does}) = 1 - .95^{20} \approx 64\%.$$

Only the product rule needs independence; the union bound does not.

Two Inequalities the Proof Will Use

For $a \in [0, 1]$,

$$1 - a \leq e^{-a} \quad \implies \quad (1 - a)^n \leq e^{-an}.$$

Thus a failure bound $me^{-\varepsilon n}$ is at most δ exactly when

$$\begin{aligned} me^{-\varepsilon n} \leq \delta &\iff -\varepsilon n \leq \log(\delta/m) \\ &\iff n \geq \frac{\log(m/\delta)}{\varepsilon}. \end{aligned}$$

The exponential comes from repeated misses; the logarithm appears when solving for the sample size n .

Finite-Class PAC Bound under Realizability

Theorem (Section 2's conjecture, made precise)

Let $S \sim \mathcal{D}^n$ be i.i.d., let $|\mathcal{H}| < \infty$, and let $\mathcal{D}$ be realizable by $\mathcal{H}$. Let $h_S \in \mathcal{H}$ satisfy $L_S(h_S) = 0$. Then for every $\varepsilon \in (0, 1)$,

$$\mathbb{P}_{S \sim \mathcal{D}^n}(L_{\mathcal{D}}(h_S) > \varepsilon) \leq |\mathcal{H}| e^{-\varepsilon n}.$$

Consequently $L_{\mathcal{D}}(h_S) \leq \varepsilon$ with probability at least $1 - \delta$ whenever $n \geq \log(|\mathcal{H}|/\delta)/\varepsilon$.

$$\text{generalization bound } \frac{\log |\mathcal{H}| + \log(1/\delta)}{n} \leq \varepsilon$$

Proof plan. Replace random h_S by fixed candidates:

bad $h_S \rightarrow$ some fixed bad rule fits $S \rightarrow$ each rarely fits $\rightarrow$ union bound.

PROOF STEP 1/3: Replace the Random Output by Fixed Candidates

Define the set of **bad** rules,

$$\mathcal{H}_{\text{bad}} = \{h \in \mathcal{H} : L_{\mathcal{D}}(h) > \varepsilon\}.$$

It is determined by $\mathcal{D}, \mathcal{H}, \varepsilon$ — **not by** S . Every member is fixed as in Case 1.

PROOF STEP 1/3: Replace the Random Output by Fixed Candidates

Define the set of **bad** rules,

$$\mathcal{H}_{\text{bad}} = \{h \in \mathcal{H} : L_{\mathcal{D}}(h) > \varepsilon\}.$$

It is determined by $\mathcal{D}, \mathcal{H}, \varepsilon$ — **not by** S . Every member is fixed as in Case 1.

Claim. $\{L_{\mathcal{D}}(h_S) > \varepsilon\} \subseteq \{\exists h \in \mathcal{H}_{\text{bad}} : L_S(h) = 0\}$.

Why. If $L_{\mathcal{D}}(h_S) > \varepsilon$, then $h_S \in \mathcal{H}_{\text{bad}}$; since $L_S(h_S) = 0$, it witnesses the right-hand event.

The right-hand event mentions no algorithm: some fixed bad rule fits S perfectly.

This step removes random h_S . The remaining proof uses only fixed rules.

In-Class Exercise: 90 seconds

Why is $\mathcal{H}_{\text{bad}}$ fixed even though h_S is random? And why does the inclusion need $L_S(h_S) = 0$?

In-Class Exercise: 90 seconds

Why is $\mathcal{H}_{\text{bad}}$ fixed even though h_S is random? And why does the inclusion need $L_S(h_S) = 0$?

$\mathcal{H}_{\text{bad}}$ is defined by $\mathcal{D}, \mathcal{H}, \varepsilon$; h_S changes with the sampled dataset. Without $L_S(h_S) = 0$, a bad h_S need not witness " $L_S(h) = 0$ " — realizability is what guarantees it.

PROOF STEP 2/3: Bound One Fixed Bad Hypothesis

Fix one $h \in \mathcal{H}_{\text{bad}}$, so $L_{\mathcal{D}}(h) > \varepsilon$. When does it make no mistake on S ?

$$\{L_S(h) = 0\} = \{h(X_1) = Y_1\} \cap \cdots \cap \{h(X_n) = Y_n\}.$$

PROOF STEP 2/3: Bound One Fixed Bad Hypothesis

Fix one $h \in \mathcal{H}_{\text{bad}}$, so $L_{\mathcal{D}}(h) > \varepsilon$. When does it make no mistake on S ?

$$\{L_S(h) = 0\} = \{h(X_1) = Y_1\} \cap \cdots \cap \{h(X_n) = Y_n\}.$$

$$\begin{aligned}\mathbb{P}_{S \sim \mathcal{D}^n}(L_S(h) = 0) &= \prod_{i=1}^n \mathbb{P}(h(X_i) = Y_i) && \text{Rule 2: independent draws; event fixed since } h \text{ is fixed} \\ &= (1 - L_{\mathcal{D}}(h))^n && \text{Rule 1: } \mathbb{P}(h(X_i) = Y_i) = 1 - L_{\mathcal{D}}(h) \\ &< (1 - \varepsilon)^n && L_{\mathcal{D}}(h) > \varepsilon; a \mapsto (1 - a)^n \text{ decreasing} \\ &\leq e^{-\varepsilon n} && 1 - a \leq e^{-a}.\end{aligned}$$

Fixed h licenses both the product rule and $\mathbb{P}(h(X_i) \neq Y_i) = L_{\mathcal{D}}(h)$. Step 1 made h fixed.

PROOF STEP 3/3 : Apply the Union Bound

Chain the three steps:

$$\begin{aligned}\mathbb{P}_{S \sim \mathcal{D}^n} (L_{\mathcal{D}}(h_S) > \varepsilon) &\leq \mathbb{P}_{S \sim \mathcal{D}^n} (\exists h \in \mathcal{H}_{\text{bad}} : L_S(h) = 0) && \text{Step 1} \\ &\leq \sum_{h \in \mathcal{H}_{\text{bad}}} \mathbb{P}_{S \sim \mathcal{D}^n} (L_S(h) = 0) && \text{Rule 3 (no independence needed)} \\ &< |\mathcal{H}_{\text{bad}}| e^{-\varepsilon n} \leq |\mathcal{H}| e^{-\varepsilon n} && \text{Step 2; } \mathcal{H}_{\text{bad}} \subseteq \mathcal{H}.\end{aligned}$$

PROOF STEP 3/3 : Apply the Union Bound

Chain the three steps:

$$\begin{aligned}\mathbb{P}_{S \sim \mathcal{D}^n} (L_{\mathcal{D}}(h_S) > \varepsilon) &\leq \mathbb{P}_{S \sim \mathcal{D}^n} (\exists h \in \mathcal{H}_{\text{bad}} : L_S(h) = 0) && \text{Step 1} \\ &\leq \sum_{h \in \mathcal{H}_{\text{bad}}} \mathbb{P}_{S \sim \mathcal{D}^n} (L_S(h) = 0) && \text{Rule 3 (no independence needed)} \\ &< |\mathcal{H}_{\text{bad}}| e^{-\varepsilon n} \leq |\mathcal{H}| e^{-\varepsilon n} && \text{Step 2; } \mathcal{H}_{\text{bad}} \subseteq \mathcal{H}.\end{aligned}$$

Make the failure probability at most δ (logarithmic inversion):

$$|\mathcal{H}| e^{-\varepsilon n} \leq \delta \iff n \geq \frac{\log(|\mathcal{H}|/\delta)}{\varepsilon}. \quad \square$$

The union bound pays once per candidate: “more candidates, more optimism” becomes $|\mathcal{H}|$. For $|\mathcal{H}| = \infty$, this bound is vacuous.

Interpreting the Finite-Class Bound

$$n \geq \frac{\log |\mathcal{H}| + \log(1/\delta)}{\varepsilon} \implies L_{\mathcal{D}}(h_S) \leq \varepsilon$$

$\log |\mathcal{H}|$

cost of choosing among
 $|\mathcal{H}|$ candidates

$\log(1/\delta)$

confidence costs only
logarithmically

$1/n$

fast rate: a consequence of
realizability

This answers “more data helps — how much?” for one setting: $n \propto \log |\mathcal{H}|/\varepsilon$. Drop realizability and the rate changes; make $\mathcal{H}$ infinite (the memorizer’s class) and cardinality stops being useful.

Try It on a Neural Network

A network with d parameters stored at b bits each has at most $|\mathcal{H}| \leq 2^{bd}$ distinct settings, so today's bound reads

$$n \geq \frac{bd \ln 2 + \ln(1/\delta)}{\varepsilon}.$$

Plug in $d = 10^6$, $b = 32$, $\varepsilon = 0.05$, $\delta = 0.01$:

$$n \geq \frac{2.2 \times 10^7 + 4.6}{0.05} \approx 4.4 \times 10^8.$$

CIFAR-10 has 5×10^4 training images.

And yet:

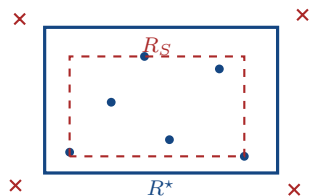
- such networks reach a few percent test error on CIFAR-10;
- the same networks fit **random labels** perfectly and then generalize at chance (Zhang et al. 2017).

The bound is valid but **vacuous**: parameter counting is the wrong complexity. Larger classes always count worse, yet test error fell. Weeks 3, 5–7, and 10 replace raw counting.

Axis-Aligned Rectangles: An Infinite Class

Problem. The threshold class is infinite, so the finite-class theorem is silent. We prove a 2D analogue; thresholds follow.

Setting. $\mathcal{H}$ contains all axis-aligned rectangles in $\mathbb{R}^2$; $Y = \mathbf{1}[X \in R^*]$ for a fixed R^* . The marginal $\mathcal{D}_X$ has a density, and $\mathcal{D}_X(B) = \mathbb{P}(X \in B)$.



Algorithm. Return the smallest rectangle R_S containing all positive training points (empty if none).

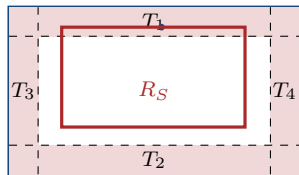
Two facts from the picture.

- 1 $L_S(R_S) = 0$: positives lie inside R_S ; negatives lie outside $R^* \supseteq R_S$.
- 2 $R_S \subseteq R^*$, so errors are only missed label-1 points: $L_{\mathcal{D}}(R_S) = \mathcal{D}_X(R^* \setminus R_S)$.

Now $L_{\mathcal{D}}(R_S)$ is the missed mass $\mathcal{D}_X(R^* \setminus R_S)$. Bound it with four fixed events.

Rectangle Bound: Boundary-Strip Argument

Trivial case. If $\mathcal{D}_X(R^*) \leq \varepsilon$, then $L_{\mathcal{D}}(R_S) \leq \varepsilon$. Otherwise $\mathcal{D}_X(R^*) > \varepsilon$:



Four fixed strips. Along the inner edges of R^* , choose $T_1, \dots, T_4$ with mass $\varepsilon/4$ each (existence: Notes, Lemma C.3). They depend on $\mathcal{D}, \varepsilon$, **not on** S .

Observation. A training point in T_1 makes the top edge of R_S reach T_1 ; similarly for the other edges. If every strip is hit,

$$R^* \setminus R_S \subseteq T_1 \cup T_2 \cup T_3 \cup T_4, \quad L_{\mathcal{D}}(R_S) \leq \sum_{j=1}^4 \mathcal{D}_X(T_j) = \varepsilon.$$

Contrapositive: if $L_{\mathcal{D}}(R_S) > \varepsilon$, at least one fixed strip was missed.

Rectangle Bound: Four Bad Events

For one fixed strip T_j of mass $\varepsilon/4$ — fixed, so Rule 2 applies:

$$\mathbb{P}_{S \sim \mathcal{D}^n}(\text{no training point in } T_j) = (1 - \varepsilon/4)^n \leq e^{-\varepsilon n/4}.$$

Union bound over the four strips (Rule 3), then the previous slide's contrapositive:

$$\mathbb{P}_{S \sim \mathcal{D}^n}(L_{\mathcal{D}}(R_S) > \varepsilon) \leq \mathbb{P}(\exists j : T_j \text{ is empty}) \leq 4e^{-\varepsilon n/4} \leq \delta \iff n \geq \frac{4}{\varepsilon} \log \frac{4}{\delta}.$$

Corollary (thresholds). For $\mathcal{H} = \{\mathbf{1}[x_2 \geq t]\}$ and true $t^* = \frac{1}{2}$, let t_S be the smallest positive training height. Errors lie in $[t^*, t_S)$. Hitting a fixed strip of mass ε above t^* gives $L_{\mathcal{D}} \leq \varepsilon$. Thus $\mathbb{P}(L_{\mathcal{D}} > \varepsilon) \leq (1 - \varepsilon)^n \leq e^{-\varepsilon n} \leq \delta \iff n \geq \log(1/\delta)/\varepsilon$.

Although $|\mathcal{H}| = \infty$, the algorithm has only four failure modes. Week 3 formalizes this idea.

Week 1 on the Error Decomposition

Under realizability and exact ERM, $\mathbf{A} = 0$ when $f^* \in \mathcal{H}$ and $\mathbf{O} = 0$. Today's bounds fill $\mathbf{E}$:

Setting	$\mathbf{A}$	$\mathbf{E}$	$\mathbf{O}$
constant rule 0 (Section 2)	$\frac{1}{2}$	0 — one candidate	0
all functions (memorizer)	0	$\frac{1}{2}$ for every n — no bound	0
finite $\mathcal{H}$, $f^* \in \mathcal{H}$ (Theorem)	0	$\leq \varepsilon$ w.p. $\geq 1 - \delta$ once $n \geq \log(\mathcal{H} /\delta)/\varepsilon$	0
rectangles / thresholds	0	$\leq \varepsilon$ w.p. $\geq 1 - \delta$ once $n \geq \frac{4}{\varepsilon} \log \frac{4}{\delta}$ (resp. $\frac{1}{\varepsilon} \log \frac{1}{\delta}$)	0

Case 1 is not a row: no rule was chosen; it gives concentration, not this decomposition.
Today: restricting $\mathcal{H}$ controls $\mathbf{E}$ but may move error into $\mathbf{A}$.

Today's Theorem Against the Five-Item Checklist

1 Object	the true risk $L_{\mathcal{D}}(h_S)$ of the hypothesis ERM returned
2 Assumptions	i.i.d. sample; $ \mathcal{H} < \infty$; realizable (Bayes risk 0); $L_S(h_S) = 0$
3 Statement	$n \geq \log(\mathcal{H} /\delta)/\varepsilon \implies L_{\mathcal{D}}(h_S) \leq \varepsilon$
4 Probability	over the draw of the training set $S \sim \mathcal{D}^n$, at least $1 - \delta$
5 Scope	uninformative when $\mathcal{H}$ is infinite (memorizer), when labels are noisy (realizability), or when $\log \mathcal{H} \gg n$ (the network)

Four steps: target = $L_{\mathcal{D}}$ above Bayes · restriction = finite $\mathcal{H}$ · cost = $\log |\mathcal{H}|/n$ · test = best-of- k , whose gap the union bound predicts.

Week 1 Conclusion

200 records speak about record 201 only through assumptions that connect the sample to the population.

A fixed rule is controlled by concentration; a rule chosen from the data also needs a bound that accounts for how many candidates it was chosen among.

Rule from Section 1

never report training accuracy

touch the test set once

more data helps

smaller model avoids overfitting

What today said

$L_S(h_S)$ is optimistic; the memorizer has gap $\frac{1}{2}$

every extra look adds a candidate: the best-of- k gap

how much: $n \geq \log(|\mathcal{H}|/\delta)/\varepsilon$ under realizability

true for counting; not the whole story (Section 1's picture)

Learning Check

A classifier has $L_S(h) = 0$ but $L_D(h)$ is unknown. Which claim is justified?

- A** $L_D(h) = 0$.
- B** Its deployment risk is small.
- C** It fits the observed sample under the stated loss.
- D** Its hypothesis class is PAC learnable.

Submit the option, confidence 1–5, and one topic to reinforce. Optional feedback may be blank.

Required fields determine completion. Correctness diagnoses learning; feedback tone never changes the mark.

Optional Practice: Ungraded

Core reconstruction

- 1 For each probability in the finite-class proof, identify what is random and its distribution.
- 2 Rebuild the proof: event inclusion $\rightarrow$ independent product $\rightarrow 1 - a \leq e^{-a}$ $\rightarrow$ union bound $\rightarrow$ solve for n .
- 3 Derive $\text{Var}(L_S) = p(1 - p)/n \leq 1/(4n)$ and apply Chebyshev.

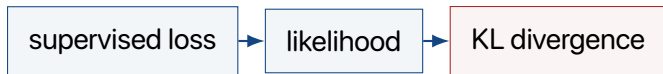
Advanced extension

- 1 Prove that absolute loss targets a conditional median.
- 2 Check each four-strip inequality and locate where density is used.
- 3 Build a no-free-lunch or high-risk interpolation example.

The notes appendices supply the prerequisite lemmas. Practice is ungraded and is not a new assessment.

Next Lecture: Maximum Likelihood and KL Divergence

What probabilistic object is a learning objective estimating?



Week 1 bounds the error of ERM in a finite realizable class. Week 2 formalizes the population target through likelihood and KL divergence.